ZEROWASTE CONNECT – SPRINT 2 COURSEWORK PRESENTED BY

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Table of Contents

Summary of Project…….

Meeting records……

User stories……

Use case diagram…….

Wireframe……

Activity diagrams…….

Kanban board screenshot…….

GitHub link……

Project Summary

Zero Waste Connect is a platform designed to reduce food waste by linking people who have surplus with those in need, Contributors can post and manage food listings, while Users can browse, search, and claim food safely. The system includes secure user login, alerts, and admin oversight to ensure smooth operation**.** This project focuses on safe, organized, and community driven food sharing, enabling contributors to manage listings and recipients to access food efficiently, all supported by secure authentication, notifications, and administrative oversight. The aim of this project is to ensure and implement features such as user authentication, role-based access control, food listing management, notifications, and administrative oversight, while building a secure, scalable, and maintainable system that facilitates community driven food sharing. This will therefore build a secure, efficient, and maintainable system using modular and well-structured code.

Meeting records

MEETING 4

meeting	Meeting 4: Sprint 2 Planning and Requirements
Date	Friday, 21 February 2026
Time	12:00 - 15:00
location	Online
attendees	Sanna, Jochebed
discussion	We began by reviewing the Sprint 2 requirements on Moodle and understanding what needed to be delivered. The main focus was on defining the application features properly and creating supporting documentation. We divided the work between us based on our strengths. Jochebed agreed to create the use case diagram showing the interactions between users and the system, the activity diagrams showing the flow of actions, and taking screenshots of our updated Kanban board. I took responsibility for writing the user stories that define our features, creating wireframes for the key pages, and documenting our meeting records. We discussed the core features that ZeroWaste Connect needs including user registration, posting food listings with photos and details, browsing available food with location-based search, and messaging

	between users to arrange collections. We also talked about the wireframes and agreed on keeping them simple and focused on the main user journeys. We planned to complete our individual tasks by the end of the week and review everything together before the submission requirement.

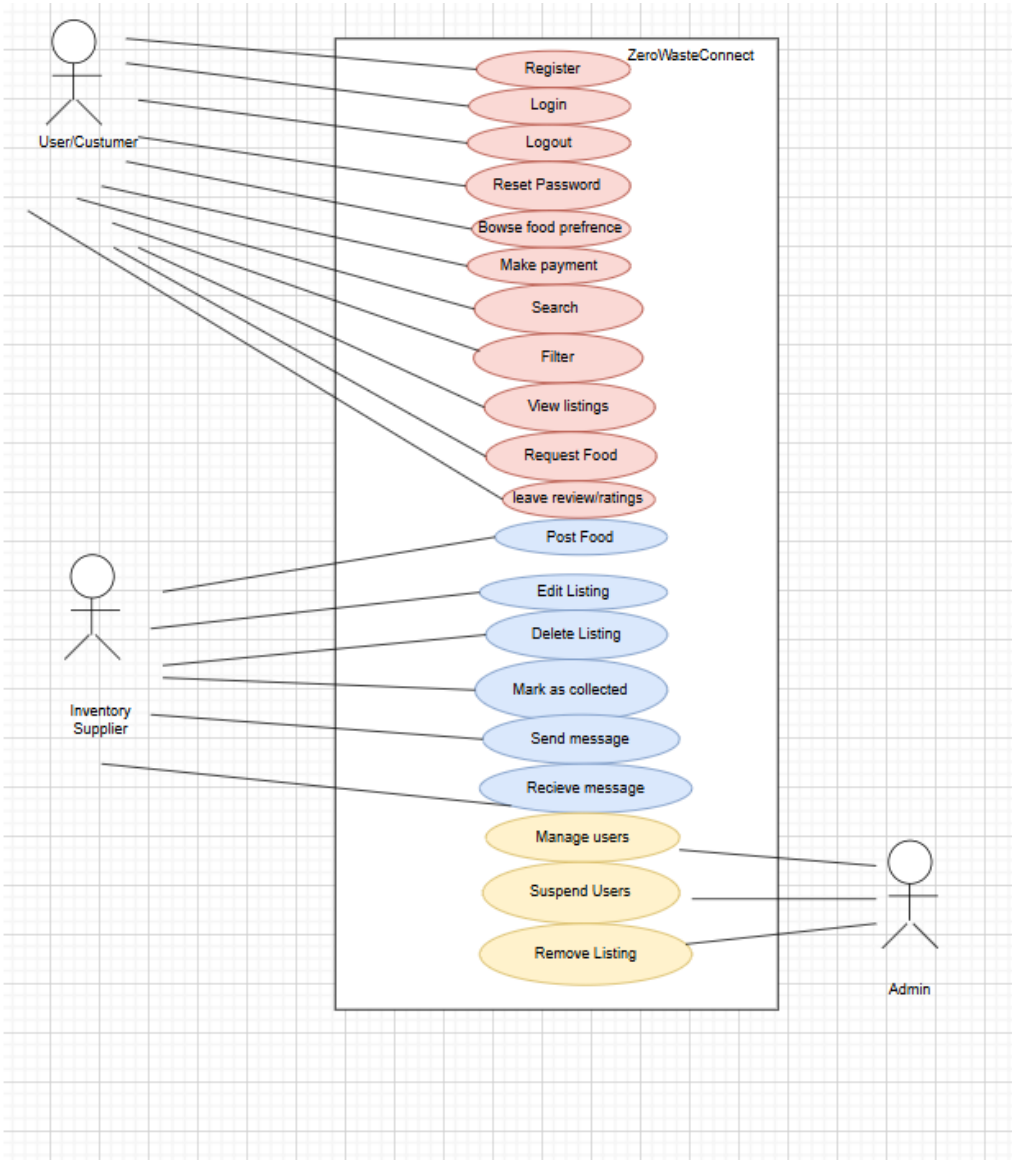
MEETING 5

meeting	Meeting 5: Sprint 2 Review and Finalisation
date	Sunday, 23 February 2026
time	12:00 - 15:00
location	**Online**
attendees	Sanna, Jochebed
discussion	We met to review all the Sprint 2 deliverables and make sure everything was complete and correct. Jochebad presented the use case diagram which clearly showed the different actors including the inventory supplier, food collectors, and system administrators, along with their interactions with the system. The activity diagrams illustrated the process flows for registering, posting a listing, and browsing food. I showed the user stories I'd written which covered all the main features we'd discussed, and the wireframes for the registration page, listing creation page, and browse food page. We reviewed them together and made some small adjustments to ensure they were clear and professional. We checked the Kanban board to make sure all Sprint 2 tasks were moved to the Done column and that Sprint 3 tasks were added to the backlog. We verified that both the GitHub repository link and Kanban board link were correct and accessible. Finally, we compiled everything into a single PDF document in the correct order as specified in the assignment brief, making sure all sections were included and properly formatted. We're both confident that Sprint 2 is complete and ready for submission.

User Stories:

- As a student, I want to post surplus food listings, so that I can share food before it goes to waste
- As a community member, I want to browse available food nearby, so that I can find free food in my area
- As a user, I want to message other users, so that I can arrange food collection times
- As a supplier, I want to upload photos of food items, so that others can see what I'm offering
- As a collector, I want to search by location and food type, so that I can find food that suits my needs close to me
- As a user, I want to rate other users after exchanges, so that I can build trust in the community
- As a registered user, I want to log in securely with my credentials, so that my account and data are protected
- As a user, I want to view my listing history, so that I can track what food I've shared
- As a new user, I want a simple registration process, so that I can start using the platform quickly
- As a food inventory supplier, I want to specify collection times and locations, so that pickups can be arranged conveniently

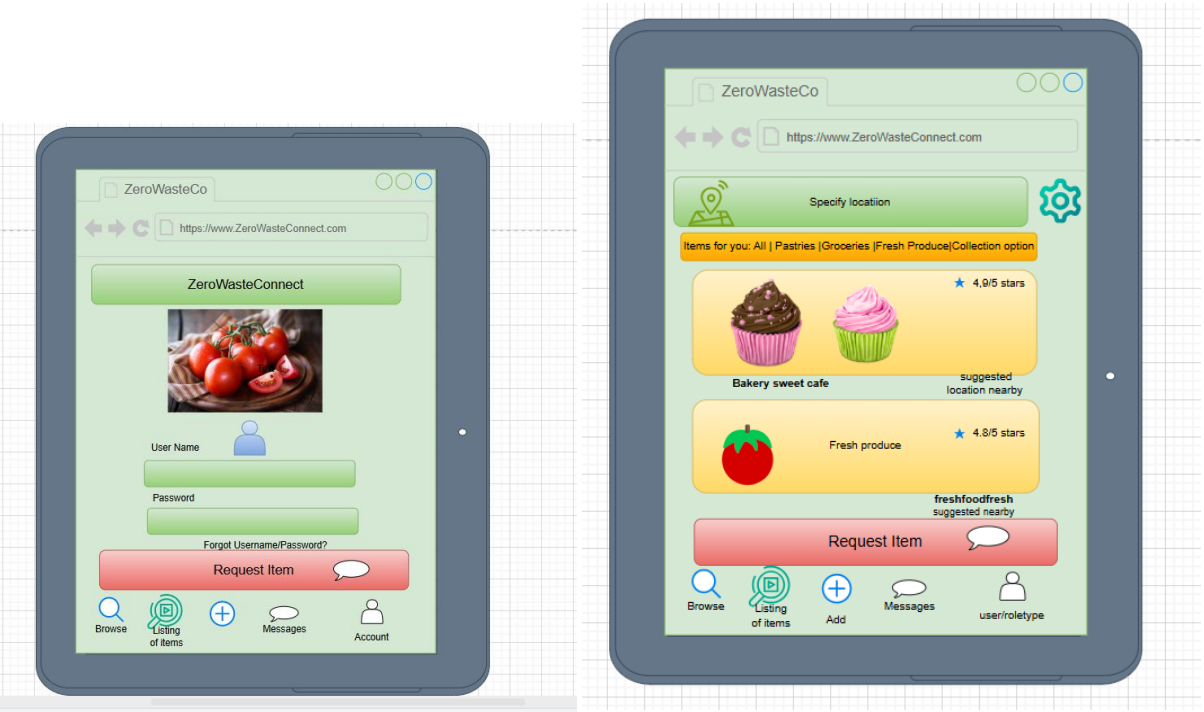
Use case diagram:



Use case diagram Specification:

In the ZeroWasteConnect system, three main actors interact within the platform which are the Administrator, the User, and the Food Provider. The Administrator, also known as the system manager or platform controller, supervises operations by managing accounts, monitoring food listings, and ensuring proper coordination of surplus distribution. The User, also referred to as a member, participant, inventory supplier, or recipient, can register, log in, post excess food items, browse available listings, request collections, and confirm pickups. The Food Provider, also called a supplier, vendor, or contributor, updates available surplus items and responds to collection requests. All interactions occur within the ZeroWasteConnect system boundary, supporting efficient food sharing, reducing waste, and strengthening community collaboration. Withal the aim of this use case diagram in to ensure that the system is designed to connect multiple user types facilitating collaboration between the inventory supplier and the user while managing operations through the admin.

Wireframes:



ZeroWasteconnect wireframes

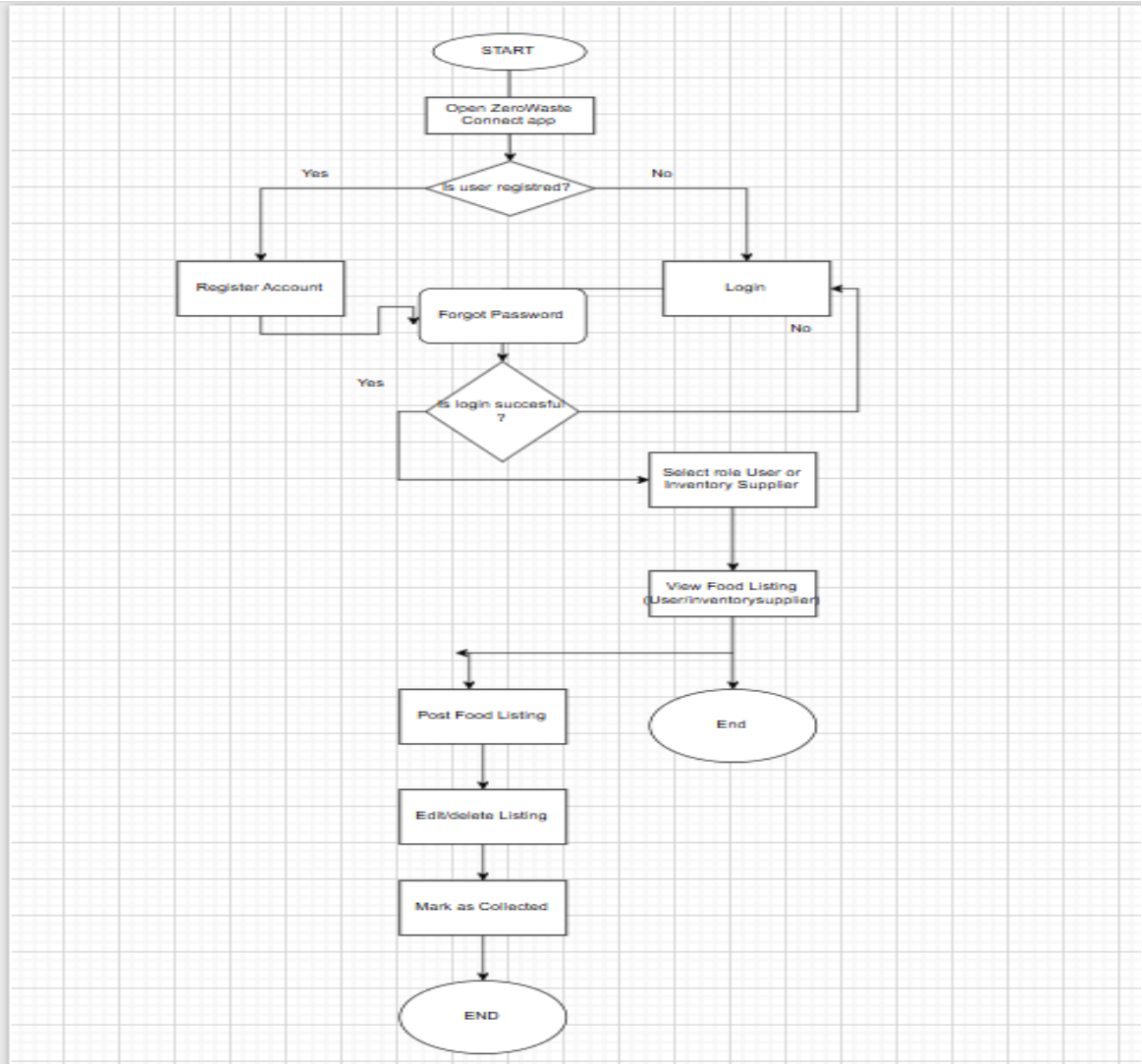
Wire frames description:

The wireframes were created to visually represent the layout and structure of the application's key screens planned for Sprint 3. They provide a low-fidelity blueprint of the user interface, focusing on functionality, navigation, and user interaction rather than visual design elements such as colour schemes or typography. Each wireframe corresponds to a specific user story and demonstrates how users will complete tasks within the system. The layouts illustrate the placement of navigation menus, input fields, buttons, content areas, and feedback messages to ensure logical user flow and usability. These wireframes help translate user requirements into a visual format, allowing early evaluation of design decisions and ensuring that the application structure supports the intended user experience before full development begins.

The wireframes illustrate the structure and organisation of the application's user interface for Sprint 3, focusing on how users will interact with the system. Each screen is designed with a clear and consistent layout to ensure usability and intuitive navigation. The interface includes a top navigation bar that allows users to move between key sections of the application, such as the dashboard, profile, and main features. Input fields are clearly labelled to guide users when entering information, and primary action buttons

Activity diagrams/flowchart

Flowchart diagram:

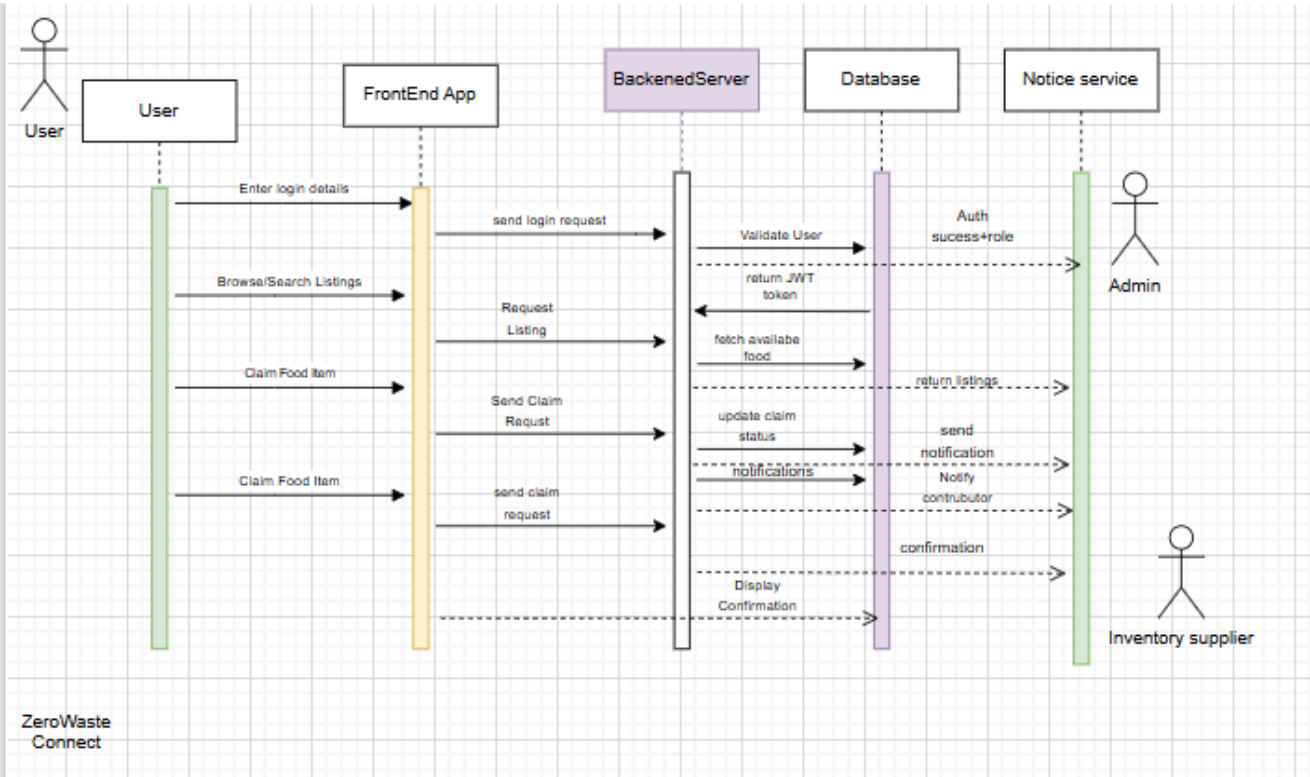


ZeroWasteConnect Flowchart diagram

Flowchart synopsis:

The ZeroWasteConnect flowchart illustrates how users and suppliers interact with the platform. Users log in securely with their account details and password and can reset forgotten. The inventory supplier can create, update, or delete food listings as shown in the flowchart. Suppliers register to contribute surplus food, manage their listings, and make them available for reservation. The flowchart shows a seamless process from authentication to inventory management and community sharing, ensuring efficient tracking and distribution of surplus food. Additionally, the platform provides real-time notifications to users when new surplus food listings become available, allowing them to reserve items promptly. Suppliers receive updates on reservations and can track which items have been claimed, helping to reduce waste efficiently. The system also includes reporting features, enabling both users and suppliers to monitor their contributions and engagements over time. By integrating these functions, ZeroWasteConnect fosters a collaborative environment where surplus food is effectively redistributed within the community.

Sequence diagram

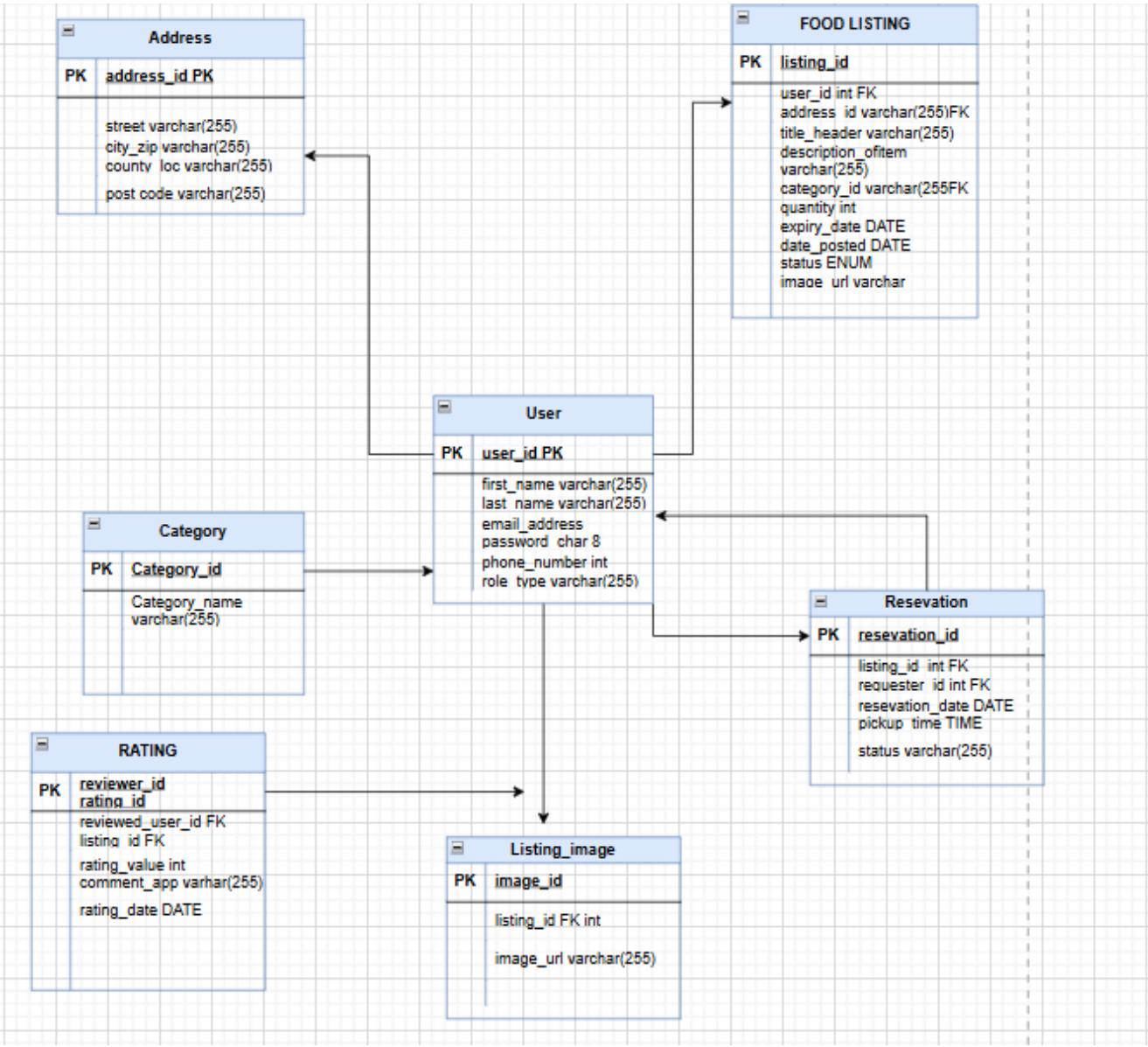


Sequence diagram demonstration:

This sequence diagram illustrates the “Browse & Claim Food” flow in the Zero Waste Connect system across five components: User/Recipient, Frontend App, Backend Server, Database, and Notification Service. The process begins with user authentication, where login credentials are sent from the frontend to the backend, validated against the database, and a JWT token is returned upon success. The user then browses listings, triggering the backend to fetch available food items from the database and return them to the frontend. When a user claims a food item, the frontend sends a claim request to the backend, which updates the claim status in the database and initiates notifications. The

notification service informs the contributor and sends a confirmation message, and finally, the frontend displays a confirmation to the user, completing the claim flow. Additionally, the sequence diagram highlights error-handling and system feedback mechanisms that ensure a smooth user experience. If a claim attempt fails due to unavailability or server issues, the backend communicates the error to the frontend, prompting the user with an appropriate message. The system also logs all interactions in the database for auditing and tracking purposes, enabling administrators to monitor usage patterns and optimize performance. By incorporating these safeguards, Zero Waste Connect ensures reliability and transparency throughout the "Browse & Claim Food" process.

ERD Diagram



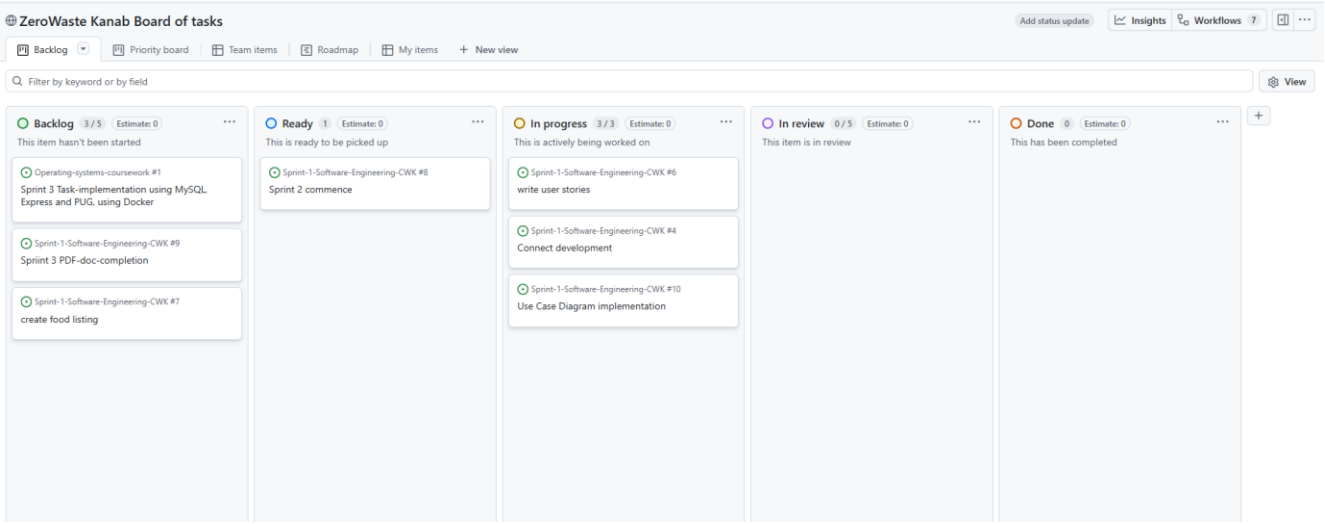
Erd Diagram illustration:

The ZeroWasteConnect ERD models the database structure of the platform, showing how users, food listings, suppliers, categories, reservations, and ratings interact. Each entity has a primary key to uniquely identify records, while foreign keys enforce relationships, such as users to listings, listings to categories, and users to reservations. Attributes use appropriate

data types, including VARCHAR for text, DATETIME for timestamps, INT for identifiers, and ENUM for controlled values like the status. Cardinality is mostly one-to-many, reflecting that a user can have multiple listings or reservations, a category can include many listings, and a listing can receive multiple reservations or ratings. This structure normalizes data, ensures integrity, and can be directly implemented in a relational database to efficiently manage users, inventory, and community interactions while supporting queries for listing

availability, reservations, and supplier contributions.

Kanban board Documentation:



Screen capture 1:

Figure 1: ZeroWaste Connect Kanban Board

Our ZeroWaste Connect Kanban board on GitHub Projects shows the progress of our development work across Sprint 2 and Sprint 3. The board is organized into five columns: Backlog (9 tasks) for work not yet started, Ready (1 task) for tasks prepared to begin, In Progress (3 tasks) for current work including Sprint 2 commencement and Use Case Diagram implementation, In Review (0 tasks), and Done (0 tasks) for completed tasks.

The Backlog contains Sprint 3 tasks including Task implementation using MySQL Express and PUG with Docker, PDF documentation completion, and creating food listing functionality. The Ready column shows Sprint 2 Software Engineering tasks ready to be picked up. Currently In Progress are Sprint 2 commencement work, Connect development, and Use Case Diagram implementation.

This Kanban board helps us track our progress, assign tasks between team members (Sanna and Jochebed), manage our workflow efficiently, and ensure we stay organized throughout

Screen capture 2:

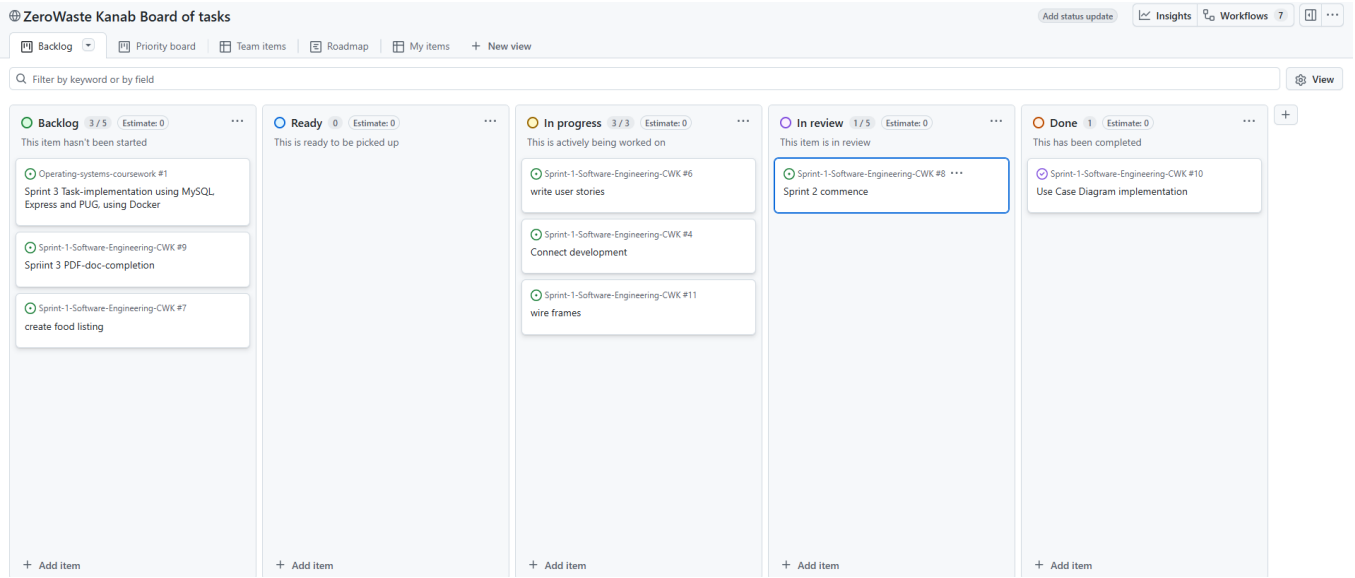


Figure 2: ZeroWaste Connect Kanban Board Detail View

The detailed view shows current task distribution. In the Backlog column, we have Sprint 3 tasks including Task implementation using MySQL Express and PUG with Docker, PDF documentation completion, and creating food listing functionality. The In Progress column contains three active tasks: writing user stories, Connect development, and wire frames creation. The In Review column has Sprint 2 commencement ready for review, and the Done column shows the completed Use Case Diagram implementation task.

This Kanban board helps us track our progress, assign tasks between team members (Sanna and Jochebed), manage our workflow efficiently, and ensure we stay organized throughout the project development process. Each task is linked to our GitHub repository and can be moved between columns as work progresses. The board clearly shows we are actively working on Sprint 2 deliverables while planning ahead for Sprint 3 tasks.

Screen capture 3:

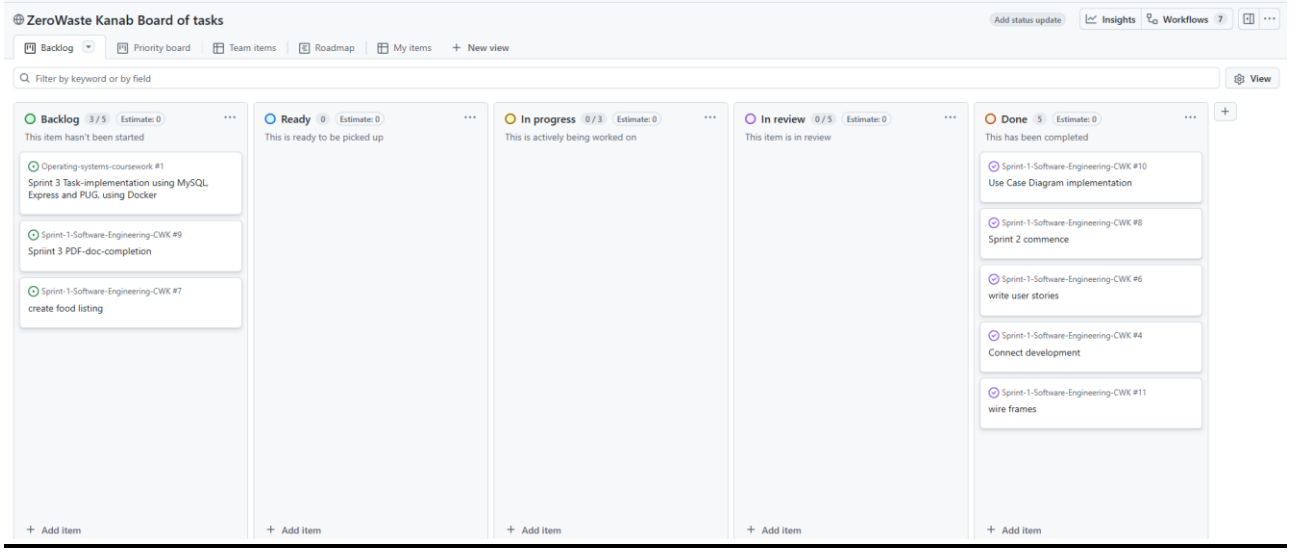


Figure 3: Illustrates the completion of the Kanban Board Task

TaskBoardLink

https://github.com/users/Joch1n/projects/3

GitHub link: https://github.com/Joch1n/Sprint-1-Software-Engineering-CWK